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An Efficient Routing Algorithm for Implementing Internet-of-Things-Based Wireless Sensor Networks Using Dingo Optimizer †

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Abstract: For Internet of Things wireless sensor networks (IOT WSNs), we suggest an energy-efficient cluster-based routing protocol. The primary issues that restrict the lifespan of a sensor network are the limited battery life of sensor nodes and ineffective protocols. Our goal is to offer a green routing protocol that wireless sensor networks can use. We present a novel approach to routing and data collection using network clustering, utilizing a modified version of the Dingo Optimizer. The main accomplishment of our suggested strategy is the elimination of the superfluous overhead with the use of cluster-head selection based on the Dingo Optimizer. Each sensor node has a data-compression method in place, which reduces the energy consumption and lengthens the lifespan of the IOT network.

Keywords: Internet of Things; wireless sensor network; base station; Dingo Optimizer; data compression

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1. Introduction

Wireless sensor networks (WSNs) have recently arisen as a data-gathering approach with a variety of utilizations. A WSN is commonly made from minimal-expense, low-power, thickly conveyed, and haphazardly dispersed sensors [1]. As well as checking the climate by taking spatial or fleeting estimations, sensors are likewise capable of steering detecting information back to an inside or outside sink [2–4]. IoT is expected to be the next major communication revolution [5]. Creating the ideal network of all wireless devices that are capable of Internet-based communication is the primary goal of the Internet of Things. An Internet of Things ecosystem is atypical due to its many diverse elements, which range in size from tiny sensors to massive, dynamic data centre nodes; its strong execution environment; and the clear nature of the data produced via intelligent phenomena [6]. Gathering physical data and turning them into useful information and labour involve a number of processes, all of which depend on the IoT ecosystem for support. In particular, certain applications require complex transformations, such as data and time-series analysis, and some are latency sensitive [7]. IoT devices, like sensors, are not very good at storing large numbers of data or sending out complex tasks. They also have limited energy and transformation support. As a result, strong components are needed to complete the transformation processes that Internet of Things applications demand. Data centers, gateways, and smart phones could be among these gadgets [8].

Resource management includes updating firmware, setting up the network architecture, and keeping an eye on performance [9]. Resource management is typically connected at the Internet of Things network end, producing all the relevant information at the user end, where device managers handle the large number of devices and data. IoT stands for Internet of Things, a development in computer technology and communication that

attempts to link intelligent items together. Anything that environments may communicate with a smart object [10]. Figure 1 shows the WSN network. One of the key challenges in reaping the benefits of this new typology is combining different technologies and concerns [11]. By obtaining environment data and surrounding information, a WSN may take centre stage. The installation of a WSN design, however, is necessary to meet the innovative challenges that have been posed online and must be evaluated before we can benefit from such considerations [12]. The primary obstacle facing the Internet of Things is the limited availability of resources, such as electricity, processing power, memory, and wireless communication range and bandwidth, all of which have an impact on routing [13].

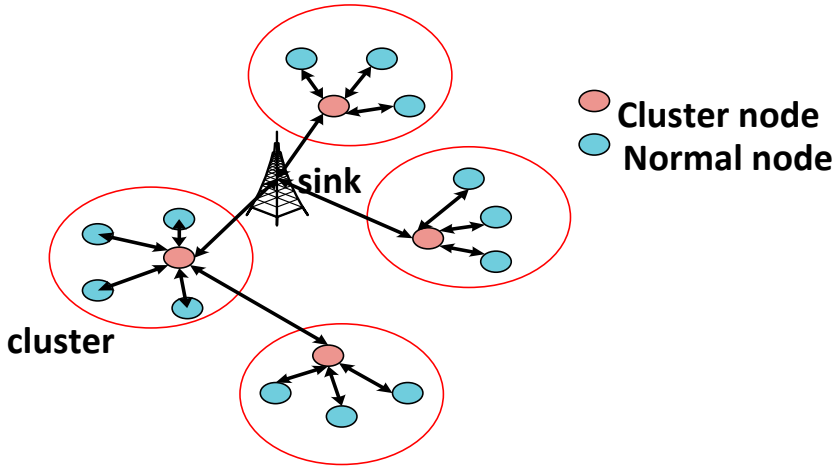

Figure 1. WSN network [1].

The availability of real devices, such as cars, buildings, and anything with actuators, sensors, implanted hardware, or programming, is made possible by the Internet of Things (IoT) [14]. These smart devices can collect and exchange data more skillfully for different purposes by enabling IoT [15].

2. Research Methodology

In summary, controlling CH energy dissipation is crucial to extending the routing protocol's network lifetime and archiving energy efficiency. Figure 2 illustrates how a WSN helps with IOT. A centroid-based assisted IoT routing strategy that enhances network behaviour has been proposed by Shen et al. [16]. The three main components of EECRP are a spread cluster evolution method that enables internal nodes to self-establish, an algorithmic sequence that modifies clusters, and spinning the cluster head (CH), located at the centre position, to evenly distribute the effective energy pack between all sensor nodes. Additionally, a new technique for reducing the effective energy utilization for high-range communication is also included. In EECRP, we consider excess energy based on nodes or by evaluating the centre position. Despite wireless sensors' exceptional short-term validity and, the element reconstruction is not worthwhile. Nevertheless, a longer routing life and better energy efficiency reaches the main goals in WSN-assisted IoT. It is true that multiple real-world objects may be able to do the same task efficiently [17].

This takes into consideration a wireless sensor node network that is dispersed at random. The position of wireless sensor nodes remains the same as the complete WSN is organized and calculated. Additionally, sensor node position data were loaded into the nodes during network establishment. It is thought that each node always knows where the base station (BS) is, as well as the rest energy. In the EECRP and CH nodes, the direct connection with the BS is taken into consideration. The primary matrix used to assess EECRP behavior is the sensor energy model. To choose the best CH nodes for clusters, a clustering algorithm is used [18]. Figure 3 shows the basic energy efficient routing protocols.

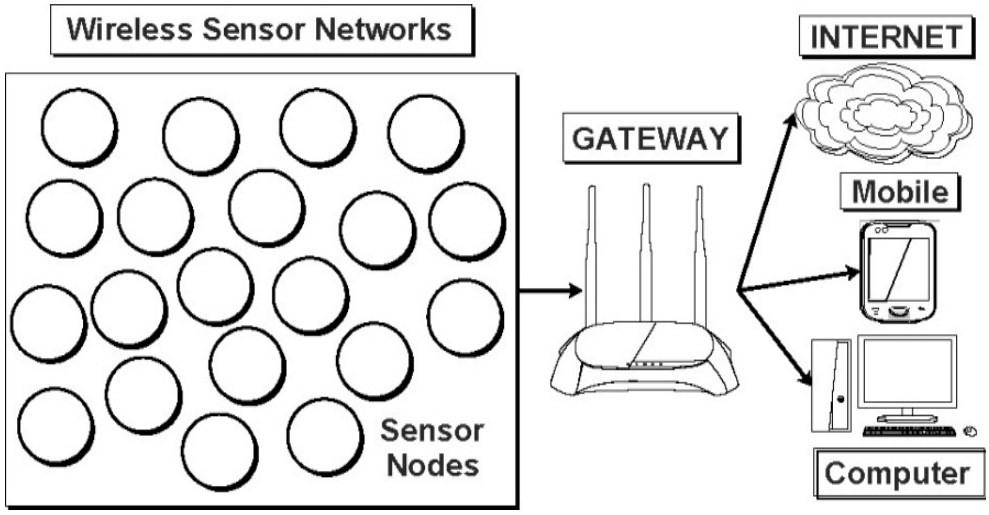

Figure 2. WSN with respect to IoT.

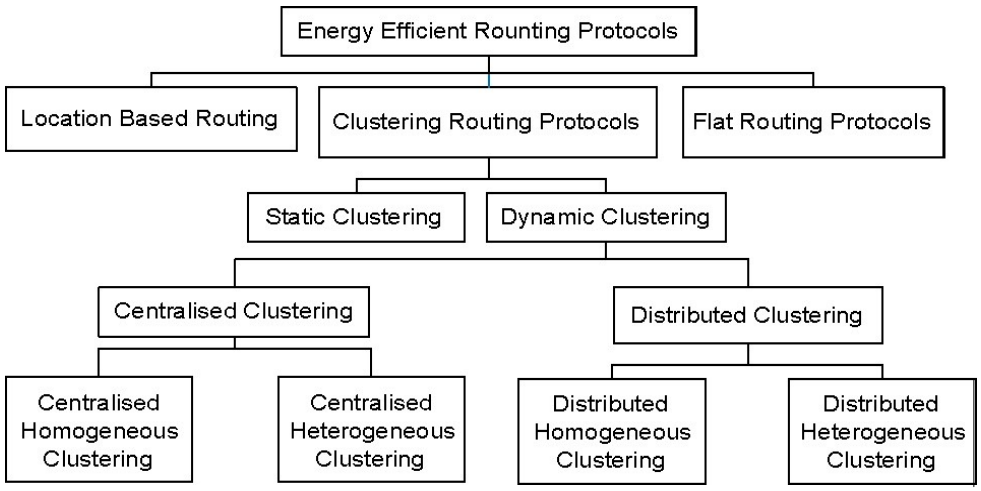

Figure 3. Energy-efficient routing protocols [2].

3. Proposed Methodology

Classical routing (CR) protocols: Because the network is homogeneous, the hard-ware complexity and battery energy of each node will be the same. Only static clustering has been applied to these networks. This approach is simple and only makes use of one network topology. All network nodes may be able to function as cluster heads, which implies that they must have the hardware capabilities to match the hardware criteria [19].

Cluster head and Relay node updating using the Dingo Optimizer: When the current CLH or RLY nodes lose energy or die, we update the CLH and RLY nodes. We apply the Dingo Optimizer algorithm to select the optimal CLH and RLY nodes in order to select the alternate CLH and RLY. The four operations that make up this algorithm are searching, encircling, hunting, and attacking prey. The following describes how this algorithm operates:

Encircling: Dingoes possess the intelligence to find their prey. Once they have tracked their prey's position, the pack and alpha surround the prey. Other search firms are in the process of revising their strategies in anticipation of this prospective approach. The subsequent mathematical Formulas (1)–(5) illustrate this behavior inspired by dingoes.

$$\vec{Z_d} = \left| \vec{U} \cdot \vec{P}_p(x) - \vec{J}(i) \right| \tag{1}$$

$$\vec{J}(i+1) = \vec{J}_p(i) - \vec{U}\cdot\vec{Z}(d) \tag{2}$$

$$\vec{U} = 2. \tag{3}$$

$$\vec{V} = 2\vec{n} * \vec{m}_2 \tag{4}$$

$$\vec{n} = 3\cdot\left(TS * \left(\frac{3}{I_{mx}}\right)\right) \tag{5}$$

The way in which dingoes can enter any location in between the points is demonstrated in detail by the random vectors *m1* and *m2*. Dingoes can move around the prey in any random spot within the quest area with the use of Equations (1) and (2). The same equations can be applied to reach a search space with N dimensions, where the dingo will go in hyper-cubes around the best outcome thus far.

Hunting: On the other hand, the theory suggests that agents usually do not compute the ideal position of the prey in the search space. We assume that the alpha, beta, and other pack members are knowledgeable about potential prey sites when we statistically develop the dingoes' hunting strategy. In this sense, Equations (6)–(11) are modelled, drawing on the discussion [20].

$$\vec{Z}_\alpha = \left|\vec{U}_1.\vec{J}_\alpha - \vec{J}\right| \tag{6}$$

$$\vec{Z}_\beta = \left|\vec{U}_2.\vec{J}_\beta - \vec{J}\right| \tag{7}$$

$$\vec{Z}_o = \left|\vec{U}_3.\vec{J}_o - \vec{J}\right| \tag{8}$$

$$\vec{J}_1 = \left|\vec{J}_\alpha - \vec{V}.\vec{Z}_\alpha\right| \tag{9}$$

$$\vec{J}_2 = \left|\vec{J}_\beta - \vec{V}.\vec{Z}_\beta\right| \tag{10}$$

$$\vec{J}_3 = \left|\vec{J}_o - \vec{V}.\vec{Z}_o\right| \tag{11}$$

The intensity of each dingo is calculated by:

$$\vec{G}_\alpha = \log\left(\frac{1}{Fit_\alpha - (1W - 100)} + 1\right) \tag{12}$$

$$\vec{G}_\beta = \log\left(\frac{1}{Fit_\beta - (1W - 100)} + 1\right) \tag{13}$$

$$\vec{G}_o = \log\left(\frac{1}{Fit_o - (1W - 100)} + 1\right) \tag{14}$$

Here, "Fit" denotes the dingo's fitness value, and "Fit_o" denotes the fitness value of additional dingoes.

Attacking Prey: The method is formulated theoretically by progressively decreasing the value of n To some extent, exploration is revealed by the suggested encircling method; nonetheless, DIOP needs more operators to emphasise exploration.

The DIOP assists its quest agents in changing their location based on the positioning of α, β, *others*, and the targeted prey. Even with these operators, the DIOP can inactivate local solutions

Searching: This is good for searching for, and avoidance of, nearby optima. Depending on a dingo's location, it will arbitrarily agree on the prey's value and make it necessary to meet the dingo rigidly. Intentionally, we used $\vec{U}.$ to provide stochastic exploration values from the initial to the final iterations. This method is effective in protecting the solution

from local optima. The Dingo Optimisation algorithm process is discussed in the following steps (Algorithm 1).

Algorithm 1: Dingo Optimization Algorithm

Input: Population of dingoes
Output: The best dingo
(1) Initial search agents Z_{in}
(2) Initialize the value of $\vec{n}$, *and* $\vec{V}$.
(3) When termination was not came do
(4) Estimate each dingo's fitness and intensity cost.
(5) Z_α= Best search with Dingo
(6) Z_β= 2nd best search of Dingo
(7) Z_o =Dingoes search
(8) repeat
(9) for
(10) $i = 1 :\ Z_{in}$ do
(11) Latest search
(12) endfor
(13) Evaluate fitness & intensity cost of dingoes.
(14) $R_\alpha,\ R_\beta,\ R_\delta$ record
(15) $\vec{n}$, $\vec{U}$ *and* $\vec{V}$ *record*
(16) $repeat = 1 + repeat$
(17)$repeat \geqslant Stopping\ criteria$
(18) output
(19) end while

4. Results and Discussion

Performance Assessment: Using the NS-2 simulation tool, the suggested routing protocol's execution will be evaluated in relation to the network's lifetime and energy consumption.

Energy Consumption: The suggested routing protocol would normalise all energy usage, which would be defined as the number of erroneous data that are placed in a network's overhead nodes for communication.

$$Energy = Power \times Time$$

Regarding the examination of energy efficiency implementation in Internet of Things devices, up to 100 nodes, the suggested system's energy efficiency will surpass that of the current system, giving the suggested algorithm the upper hand. In contrast, the suggested approach computes preventive measurements and assesses the network lifetime as a whole equally [19,20].

CH nodes with an average transmission data network are shown in Figure 4, which makes it evident that the data integrity was low. When there are a lot of CH nodes in a sensor network, the sensor will always be in bad condition because the sensor's lifetime will be shorter, data transmission will be slower, and energy dissipation will quickly cause a shortage in the network. Figures 5–7 determine the efficiency in allocating the sensor nodes and analysing the energy consumption and how to utilise the node, and determine the throughput and average delay for better performance [21–25].

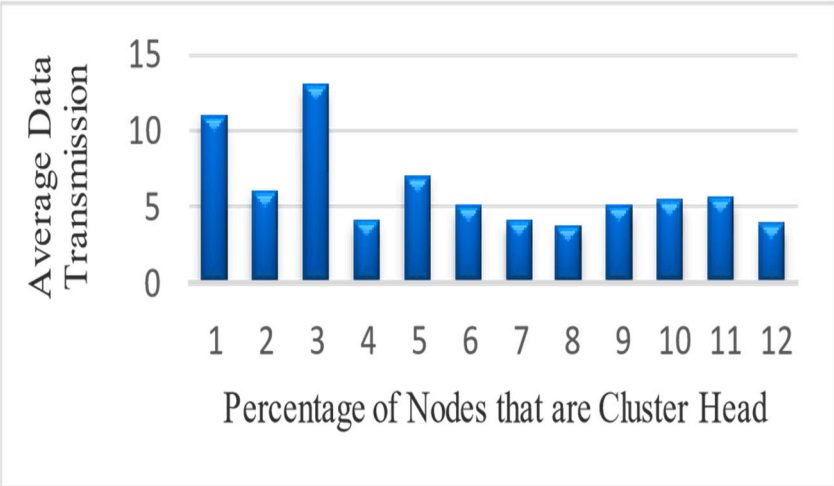

Figure 4. Different transmission data values.

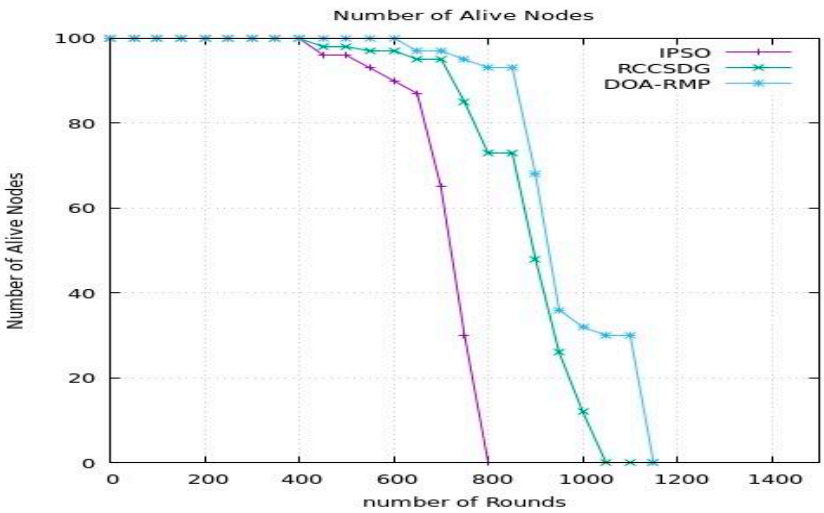

Figure 5. Average delay.

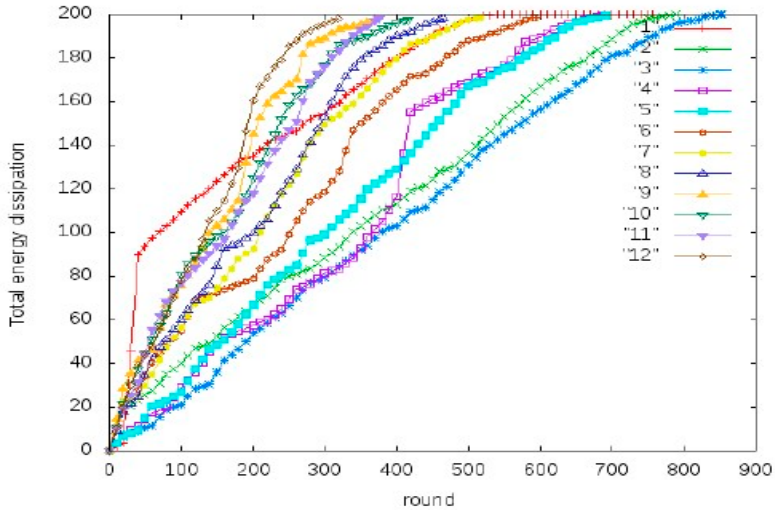

Figure 6. Energy dissipation with different values.

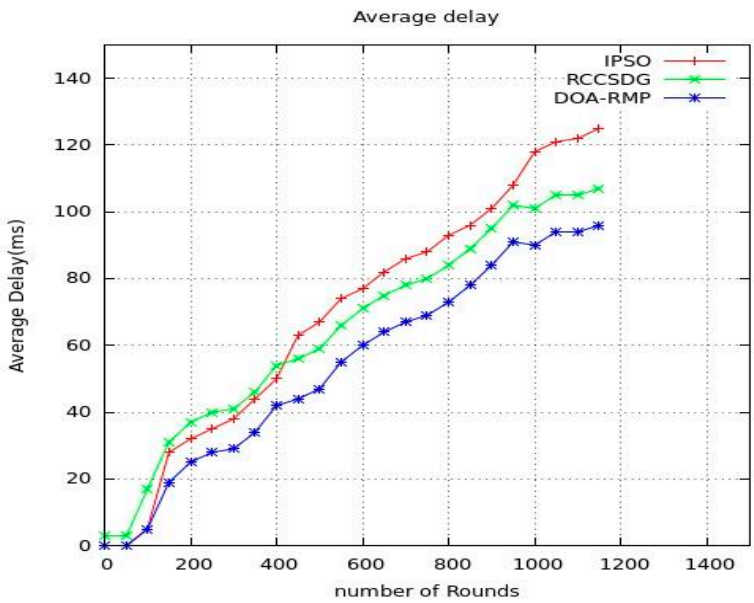

Figure 7. Average delay using the Dingo Optimizer.

5. Conclusions

One of the most important resources for WSNs is energy. The majority of studies in the literature on WSN routing have highlighted energy conservation as a crucial optimisation objective. To successfully extend the network lifetime, however, merely saving energy is insufficient. Performance is negatively impacted by uneven energy depletion, which frequently leads to network partitioning and low coverage ratios. In contrast to conventional wired networks, energy saving in wireless sensor networks has garnered significant attention recently and presented some specific issues.

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